5. System Response and LTI Properties

Objective

This experiment is divided into four simulations:

1.Impulse Response of the System

2. Step Response of the System

3. Verification of Linearity Property of LTI Systems

4. Analysis of Discrete-Time System

Experiment 5.1

Impulse Response of the System

Aim:

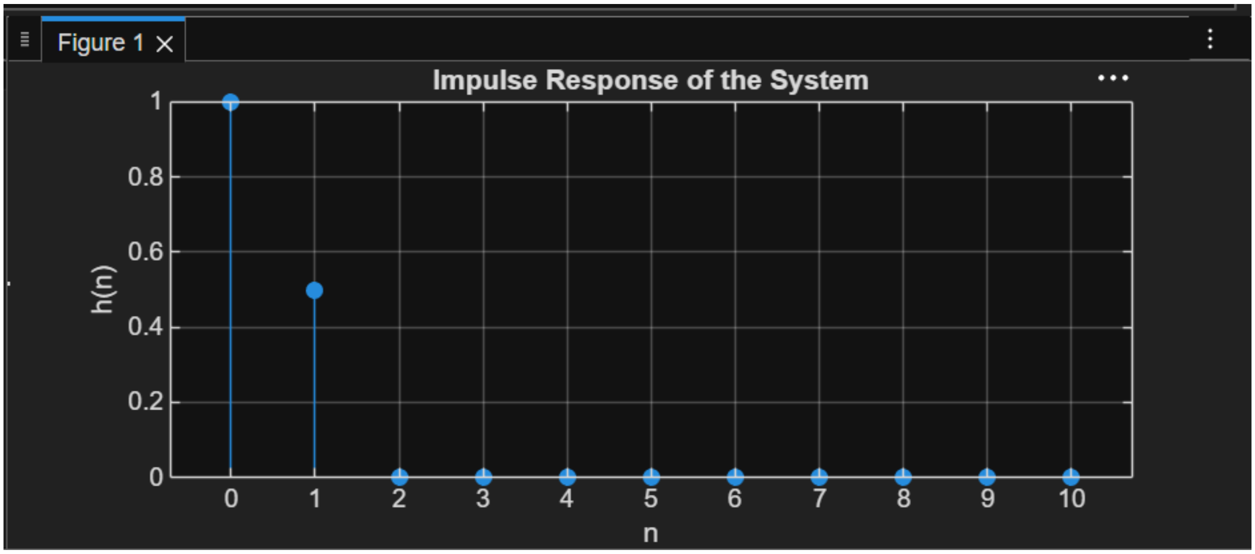
To determine and plot the impulse response of a continuous-time system. This involves applying an impulse function δ(t)\delta(t)δ(t) as the input and observing the system's output.

Program:

```
% Impulse Response of the System

clc;

clear;

close all;

% Sample index

n = 0:10;

% Unit impulse input

x = [1 zeros(1,10)];

% System: y(n)=x(n)+0.5x(n-1)
```

```
b = [1 0.5];
a = [1];
% Impulse response
h = filter(b,a,x);
% Display result
disp('Impulse Response h(n):');
disp(h);
% Plot
figure;
stem(n,h,'filled');
grid on;
title('Impulse Response of the System');
xlabel('n');
ylabel('h(n)');
```

Output Waveform:



Result:

The impulse response of the discrete-time system y(n)=x(n)+0.5x(n−1) was obtained by applying a unit impulse signal as input. The resulting impulse response was h(n)={1,0.5,0,0,...}, which characterizes the behavior of the system.

Experiment 5.2

Step Response of the System

Aim:

To analyze the step response of a discrete-time system. The step response characterizes how the system responds to a step input, which is a common way to assess the transient and steady-state behavior of the system.

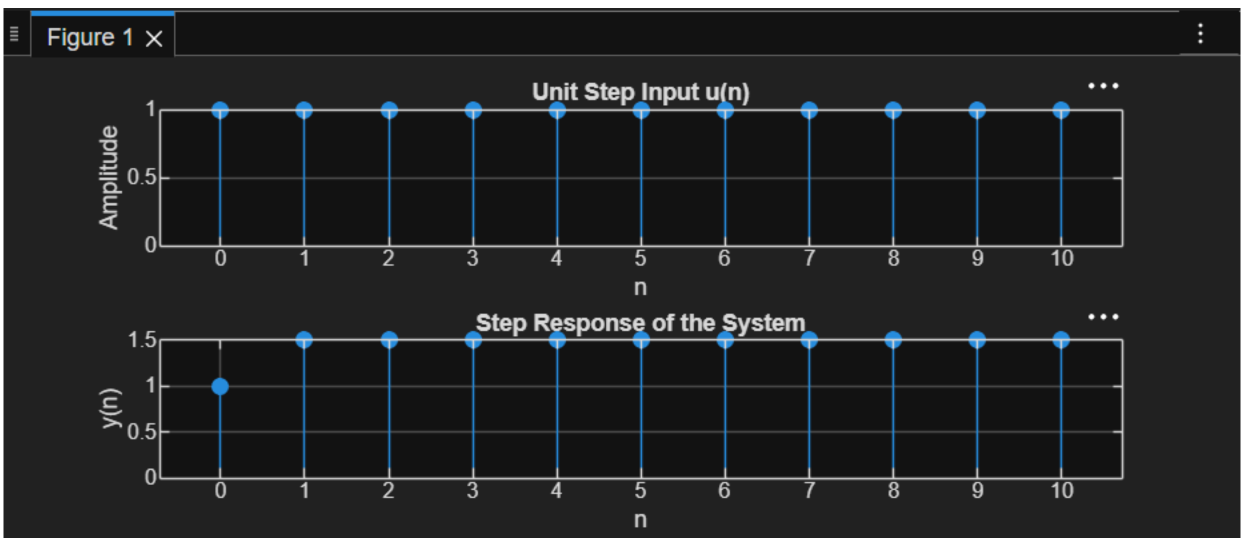
Program:

```
% Step Response of the System

clc;

clear;

close all;
```

```
% Sample index
n = 0:10;
% Unit step input
x = ones(1,length(n));
% System: y(n)=x(n)+0.5x(n-1)
b = [1 0.5];
a = [1];
% Step response
y = filter(b,a,x);
% Display result
disp('Step Response y(n):');
disp(y);
% Plot
figure;
subplot(2,1,1);
stem(n,x,'filled');
grid on;
title('Unit Step Input u(n)');
xlabel('n');
ylabel('Amplitude');
subplot(2,1,2);
stem(n,y,'filled');
grid on;
title('Step Response of the System');
```

```
xlabel('n');
ylabel('y(n)');
```

Output Waveform:



Result:

The step response of the discrete-time system y(n)=x(n)+0.5x(n−1) was obtained by applying a unit step input. The output reached a steady-state value of 1.5, demonstrating the response of the system to a constant input signal.